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How to Challenge Your Own Analysis So Others Won't

Master the art of sanity checks to level up the quality of your work



Torsten Walbaum · Follow

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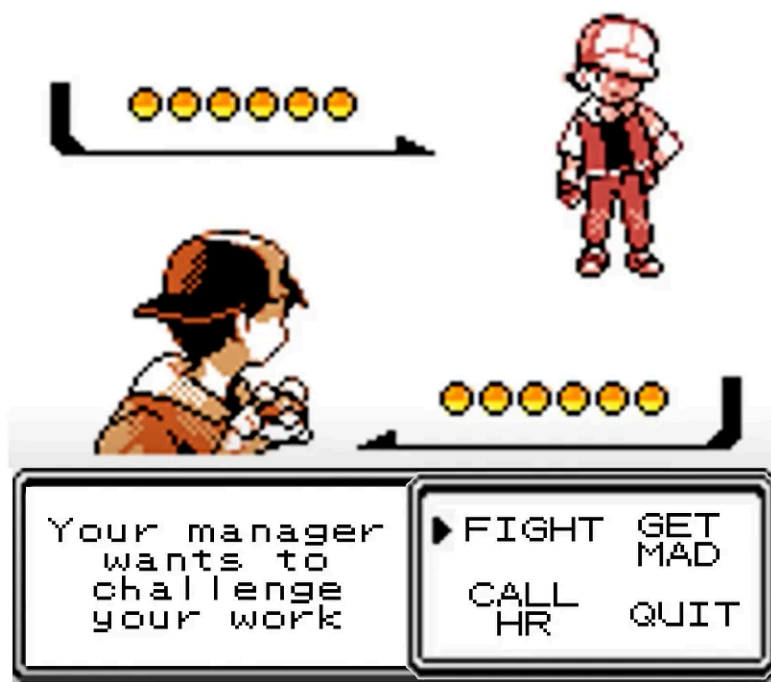


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Have you ever created an analysis only for it to be torn apart by your manager? Or have you ever gotten a question during a presentation that made you think “Why didn’t I check this beforehand?”

Sometimes it can feel like managers and executives have an uncanny ability for finding the *one* weak spot in your work. How did they identify the issue so quickly, especially if they are seeing your work for the first time?

What seems like a superpower can be learned by anyone, and this post will show you how.

By routinely applying “sanity checks” to your work, you can proactively identify the weak spots and ensure the result makes sense before you share it with a broader audience.

I am going to go over:

- What sanity checks are and why they matter
- How sanity checks are different from how most people check their work
- How to do a sanity check
- How to use sanity checks to increase your credibility
- How to use AI to sanity check your work for you

We’ve got a lot to cover, so let’s jump in.

What is a sanity check, and why is it important?

Imagine you’re building a detailed model from scratch, carefully choosing each assumption and combining them to get to your final output (e.g. a forecast, company valuation etc.).

Each assumption seemed reasonable and you checked the math twice, so the output should be solid. Right? Right??

My experience over the last decade has been that more often than not, we miss the forest for the trees when building models or doing analyses. We layer so many assumptions on top of each other that the final result can quickly change from reasonable to ridiculous.

This is where sanity checks come into play: *Sanity checks help us determine whether the result of our analysis falls has a good chance of being correct.*

We're all wrong from time to time. That's fine; reality doesn't always play out the way we expect. But you should try to be right *most of the time*.

Let's dive into how to do that.

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How is sanity checking different?

When checking our work, most of the time we go through it step-by-step to check for errors. Are the cells linked correctly? Did I pull the formulas all the way down? Do all the joins in my SQL work correctly?

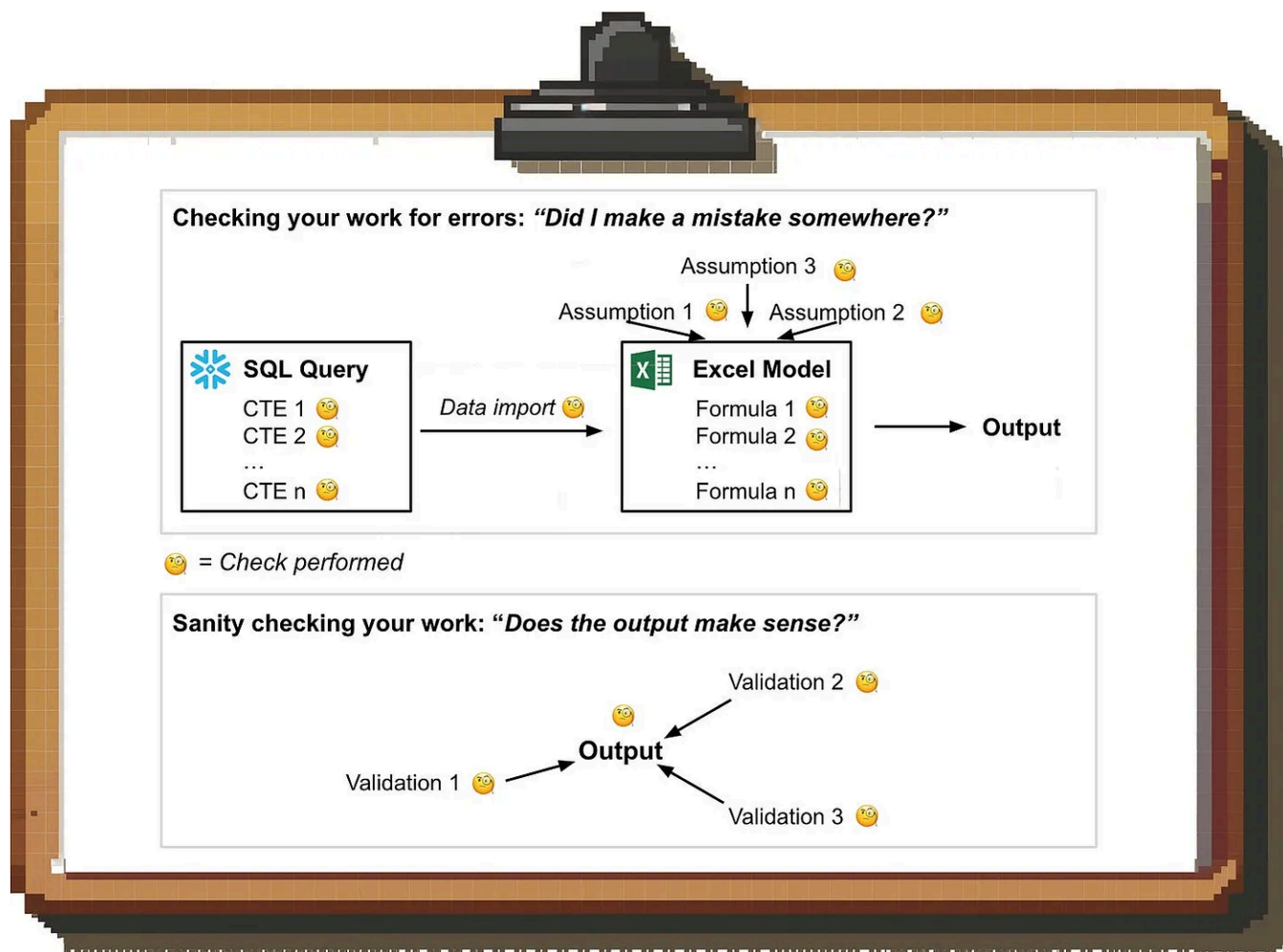


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This mechanical “Quality Control” approach of checking all inputs can help us find issues, but it doesn’t ensure that the output makes intuitive business sense.

Sanity checking, on the other hand, *is about taking a step back and validating the output from a different angle*. If you come to the same conclusion both ways, you can be much more confident in your work.

How to do a sanity check

There are three broad categories of sanity checks: Bottom-up vs. top-down, benchmarking and intuition. I will go through each of them in detail and show how you can apply them at work.

Bottom-Up vs. Top-Down

Our analyses are typically either top-down *or* bottoms-up. But what does that mean?

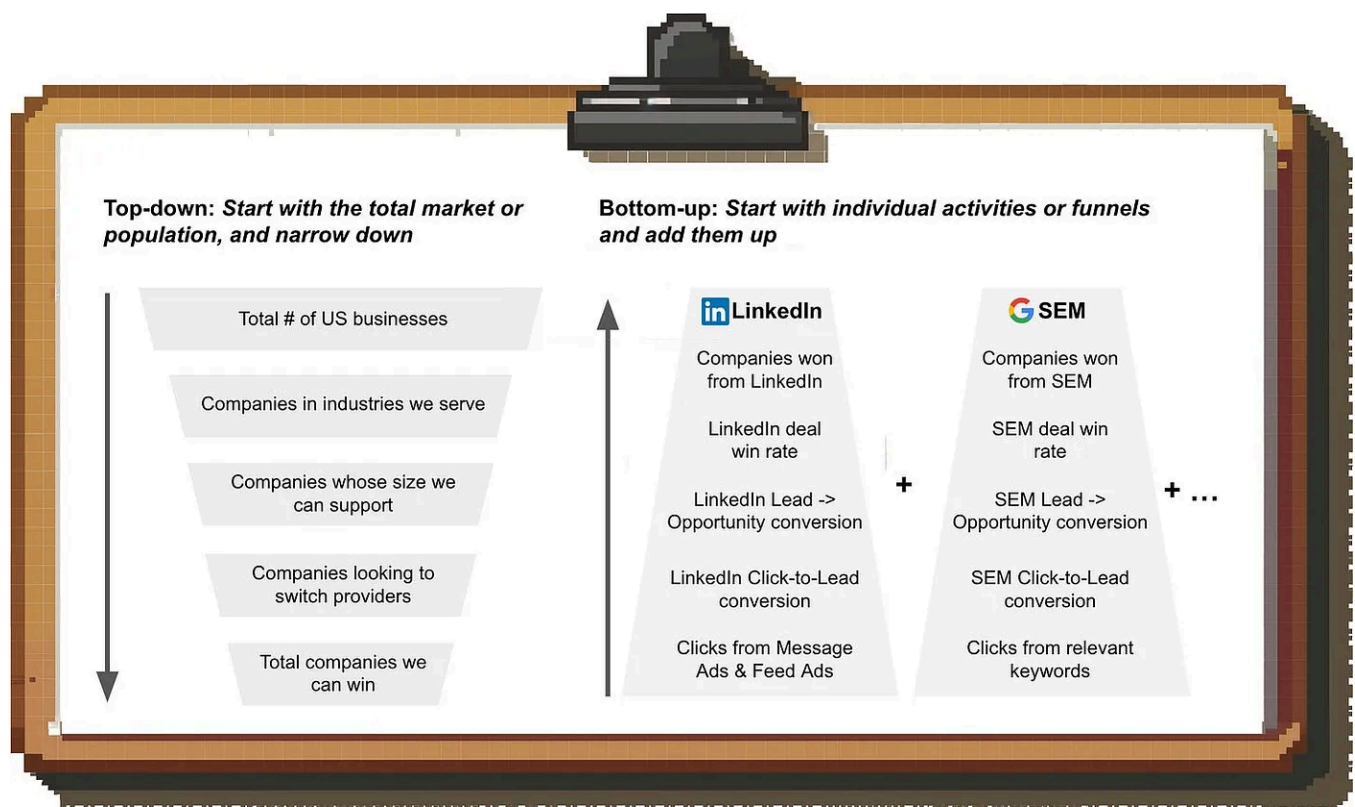


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Let’s look at a (simplified) example. Let’s say you work in a B2B SaaS company and want to figure out how many customers you can acquire for the new product you’re launching.

- In the *top-down* approach, we are trying to understand what share of the market we can win. So we’d start by looking at the total number of businesses in the US, exclude industries that we are not targeting and company sizes we can’t support,

assume what % of companies is looking to switch software providers and finally assume what share of those companies we can win (vs. our competitors)

- In the *bottom-up* approach, we are trying to understand how many companies we can acquire based on the channels we have available. So we'd look at prior launches to figure out what lead volume we can get from LinkedIn, analyze keywords to determine expected SEM volume, project Email leads based on the number of companies we can target and expected conversion rates etc.

Both approaches can give us a directional idea of what we can expect, **but they each have a crucial weakness.**

The top-down approach does not consider *how* we are going to get these customers, while the bottom-up approach ignores the size of our target market.

As a result, the best way to sanity check your work is to combine top-down with bottom-up analysis.

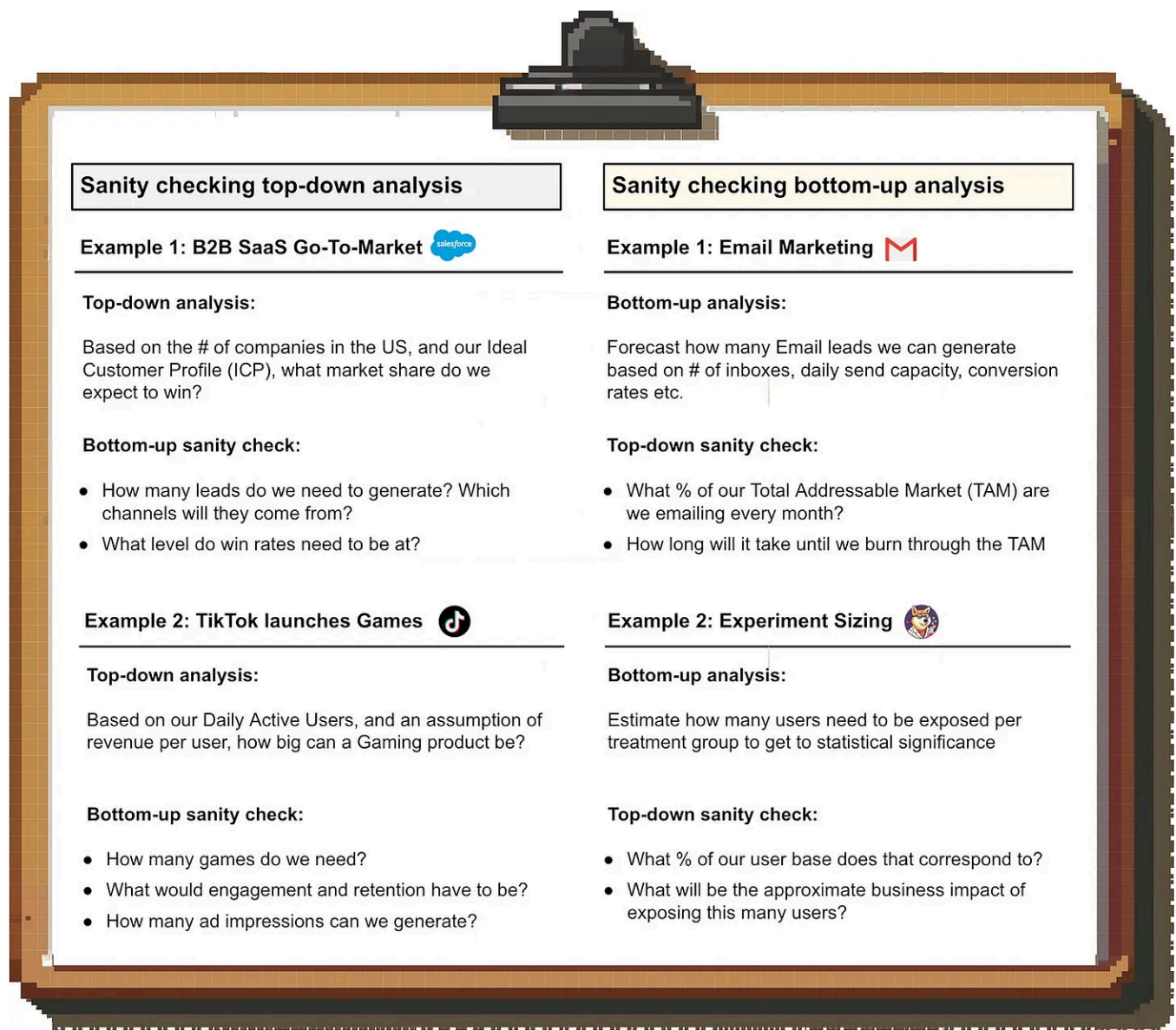


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Benchmarking

The best way to make sure a plan or projection is reasonable is to compare it against benchmarks. For example, if you're forecasting the performance of a new market, it helps to compare it against prior launches of similar countries.

If your analysis massively deviates from the benchmark, you need to be able to explain why.

A few common things you should check for any model, forecast or projection:

- **Magnitude:** How do the final outputs compare to benchmarks? E.g. are you projecting that France will be a larger market for the company than the UK?
- **Growth assumptions:** What trend are you forecasting over time? E.g. is the new product projected to grow more quickly compared to past launches?

- **Seasonality:** Does your projection show the same repeating patterns as the benchmarks? E.g. if all other markets show a slowdown during the December holiday period, why does your projection for the new country not show this?

This doesn't mean you always have to model everything in line with benchmarks; but you always need to be able to explain why something is deviating.

Benchmarking Example 1: New Market launch

Scenario: You plan to enter the UK market and forecast user growth

✅ **Sanity checks:** You compare the forecast against the two most recent launches, Germany and France. Your forecast is more aggressive than the last two launches, and doesn't show any seasonality.

? Questions you need to be able to answer:

- What gives you confidence that that's possible? Is the UK structurally different (e.g. the market is larger)? Is our product a better fit for the UK market? Are we using a different go-to-market strategy?
- Why is there no seasonality in France? Are holiday periods different? Do B2B buyers have different seasonal buying patterns?

👉 If you are not able to make a strong case for why the new market is different from past launches, you are better off keeping the forecast similar.

Benchmarking Example 2: Marketing Plan

Scenario: You are forecasting Marketing spend and performance metrics by channel. The plan is to double the Marketing budget year-over-year.

✅ **Sanity check:** You compare the projected marketing efficiency against past trends. Your forecast projects that efficiency (Cost per Lead) will improve as we increase Marketing spend, but past data shows the opposite trend.

? Questions you need to be able to answer:

- Why do you expect better efficiency?
- What specific improvements are we deploying in each Marketing channel that will drive this?

- Are we doing anything that would improve overall Marketing performance, e.g. investing in our Brand?

👉 If you don't have a concrete plan to improve Marketing efficiency, you should assume that the historical relationship between spend and efficiency holds.

Comparing against intuition

Many times, you can use common sense to sanity check your analysis. Your intuition is not always going to be right, but it often highlights potential issues that need further validation.

A few examples:

- You build a Discounted Cash Flow financial model and the terminal growth rate is much higher than that of the Gross Domestic Product (GDP); this means you are *implicitly assuming that the company will outperform the broader economy forever*. Is that realistic?
- You are building an Account Scoring model and it scores companies as “good fit” that Sales believes are a waste of time. This doesn't mean Sales is right (you build the model to surface new insights, after all), but you should take their experience into account since *counterintuitive outputs often highlight model weaknesses*

Using sanity checks to increase your credibility

Sanity checks are not just there to prevent others from poking holes in your work. They also give you a tool to increase your credibility.

Instead of doing them behind the scenes and then sharing the improved work, share the sanity checks as well. By showing how you validated the output of your analysis, you will build trust with your audience. *If you don't share the sanity checks you did, the audience has to assume they have to check your work on the spot.*

You can do this visually on a slide by showing how both top-down and bottom-up approaches get to the same result, or comparing your data against benchmarks.

But you can also do it verbally:

✅ “We are planning to grow to 50 Email leads in the UK by October; that is based on similar conversion assumptions as in Canada, and corresponds to 3% monthly

How to use AI to sanity check your work

Sanity checking can be pretty time consuming; after all, you have to approach the same problem from multiple angles. Luckily, AI tools can save you a lot of time.

This is not a replacement for your sanity checking skills; ChatGPT needs your guidance to do a good job, so you still need to know how to perform a robust sanity check. The AI's job is simply to do the heavy lifting for you, and to bring up a few points you might have missed.

Here is a step-by-step guide on how to do this with ChatGPT; all screenshots are from actual conversations I had where I asked ChatGPT to sanity check my forecast for a new market launch.

Disclaimer: Always check your employer's policies on using AI tools like ChatGPT before uploading any proprietary data.

Step 1: Upload your work to ChatGPT

The first step is to upload the work that you want ChatGPT to sanity check. ChatGPT can handle a variety of file types, including PDFs, Excel, CSV files and more.

You can also integrate directly with several tools; e.g. in this example, I linked my Google Sheet that contained my forecast:

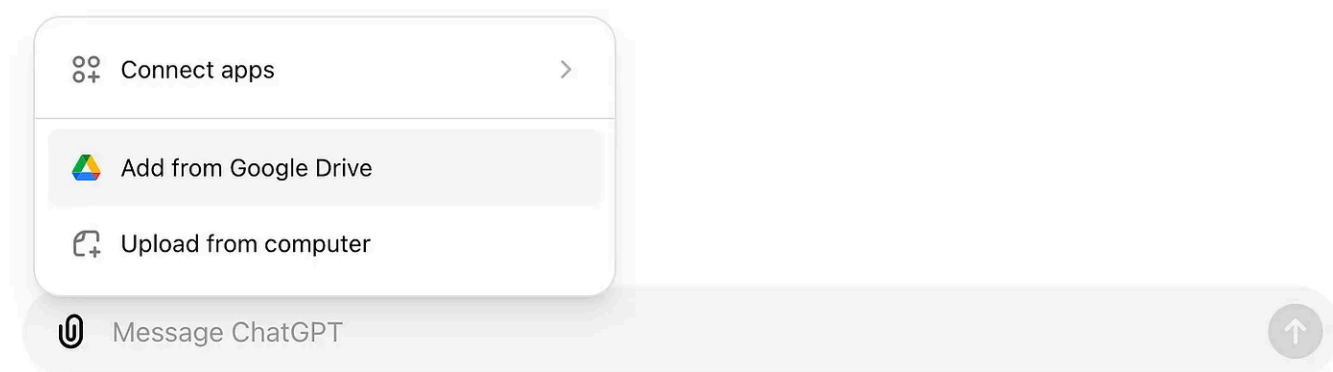


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Even if your actual model lives outside a spreadsheet (e.g. in Python), I recommend dumping the outputs in Google Sheets for the sanity check; after all, you want ChatGPT to validate your outputs, not the mechanics of your model.

For this example, I gave ChatGPT this simple Go-To-Market Forecast for a new country launch (you can make a copy and try your own sanity check with it).

I went through four sanity checks; here are the logs:

1. First attempt (Grade: Intern)
 2. Second attempt (Grade: Intern)
 3. Third attempt (Grade: First-year analyst)
 4. Fourth attempt (Grade: Over-confident first-year analyst)
- Attempt #1 and attempt #2 were okay, but I felt like I had to provide quite a lot of guidance and didn't always get exactly what I wanted.
 - The third attempt was pretty good, but both ChatGPT and I forgot to dig into the Marketing channel mix (and when I remembered a few days later, ChatGPT was unable to continue where we left off).
 - Attempt #4 was promising, but when re-forecasting to include seasonality, it adjusted the wrong months at first.

You might have to try a few times until you get a really good performance. And remember:

Don't blindly use anything that AI produces; it can (and will) make mistakes, and you'll be on the hook for the end result. AI can give helpful input and save you time, but it's not a replacement for critical thinking.

Step 2: Write a prompt asking ChatGPT to sanity check

After ingesting your file, you need to write a prompt asking ChatGPT to sanity check your work.

Here's what I used to sanity check the Go-To-Market Plan I linked above:



Prompt to Chat GPT:

Some context on the data:

Our business is a B2B SaaS company selling payroll software to small and medium sized businesses. Sales Qualified Opportunities (SQOs) are opportunities that have been generated by Marketing and accepted by the Sales team to enter the Sales funnel.

We started in the USA but have recently expanded in to other countries. France is our latest launch (launching this month), and we are forecasting SQOs for France.

My request to you:

Please sanity check the forecast for France. Make sure the forecast is reasonable; this involves comparing across the various countries in the data, but also pulling in additional data points you have access to to validate the assumptions. I want you to look at the forecast from all possible angles to make sure I didn't make a mistake in my projections.

Please highlight anything that seems odd / overly conservative / overly aggressive with regards to France so I can take a closer look.

Image by author

I've found that giving a little bit of context on the dataset is helpful (although not absolutely necessary). Also, don't forget to say "please" and "thank you" in case the AI ever gets sentient; this will keep you off the naughty list.

You could also give ChatGPT this article, or another summary of how to do sanity checks, so you don't have to include too many instructions in your prompt.

Step 3: Make sure ChatGPT ingested the data correctly

After ingesting the file, ChatGPT will often typically give you a brief summary of what it sees and what it believes the data represents, as well as the steps it will take for the analysis:



ChatGPT Output:

The Excel file contains the following sheets:

1. **France Marketing SQO Forecast**
2. **SQO Split by Marketing Channel**
3. **Launch Dates**

I'll start by reviewing the "France Marketing SQO Forecast" to understand the forecasted SQOs for France. Then, I'll compare this data with prior launches using the "Launch Dates" and "SQO Split by Marketing Channel" sheets.

Let's begin by examining the "France Marketing SQO Forecast" sheet. [-]

The "France Marketing SQO Forecast" sheet includes monthly forecast data for several countries, including France. It spans from January 2021 to June 2025.

To evaluate the forecast for France, I will:

1. Compare the initial monthly SQOs for France with those for other countries at the time of their respective launches.
2. Analyze the growth trajectory for France compared to other countries over similar periods.

Next, I'll extract and compare the launch data for each country. Let's review the "Launch Dates" sheet to understand when each country was launched. [-]

The "Launch Dates" sheet provides the launch dates for each country:

- USA: January 1, 2021
- Canada: March 1, 2022
- Australia: February 1, 2023
- UK: June 1, 2023
- France: June 1, 2024

Screenshot from my [third attempt](#); Image by author

ChatGPT will also often restate some of the data in the chat, which is helpful for making sure it pulled the numbers correctly:



ChatGPT Output:

Let's compare the initial and subsequent monthly SQOs for each country from their respective launch dates. This will help us evaluate the forecast for France against historical data from other launches. [-]

Post-Launch SQOs Comparison				
		USA	Canada	Australia
7	Month 7	115	56	44
8	Month 8	105	71	49
9	Month 9	135	85	54
10	Month 10	160	70	61
11	Month 11	194	100	40
12	Month 12	165	110	57

Screenshot from my [third attempt](#); Image by author

Note: Initially, ChatGPT had some problems ingesting my spreadsheet. **Here's how you can troubleshoot:**

- If ChatGPT throws an error or obviously didn't ingest the data correctly, you can ask it to reprocess by clicking on the  icon
- If it repeatedly has issues with your file, then you might have to clean it up. For example, I found that I could significantly reduce errors if I removed non-essential rows and columns (e.g. section headers, comments etc.). If the file only contains the relevant tables, it's easier for ChatGPT to convert them to data frames in Python. Giving descriptive column and row headers also helps ChatGPT make sense of the data

Step 4: Work through the sanity checks with ChatGPT

Next, ChatGPT will start providing some initial observations, like this:



ChatGPT Output:

Observations:

1. Initial Forecast for France:

- The initial forecast of 2 SQOs for France in the first month seems quite conservative compared to other countries' initial month SQOs.

2. Growth Rate:

- The growth trajectory for France is comparable to the UK but starts much lower. It ramps up significantly from Month 1 to Month 12, indicating a potentially aggressive growth expectation after a conservative start.

3. Comparative Analysis:

- France's growth from 2 to 41 SQOs within a year seems reasonable in comparison to the UK's growth from 11 to 41. However, the initial number is significantly lower, which might suggest either a very cautious initial forecast or an underestimation of the French market's potential.

Screenshot from my [third attempt](#); Image by author

ChatGPT correctly identified that the forecast is conservative and catches up with the UK eventually after a slow start. It would have been better if it had pulled in some stats for validation such as the number of small businesses in each country, but in my experience you need to ask it explicitly to do that.

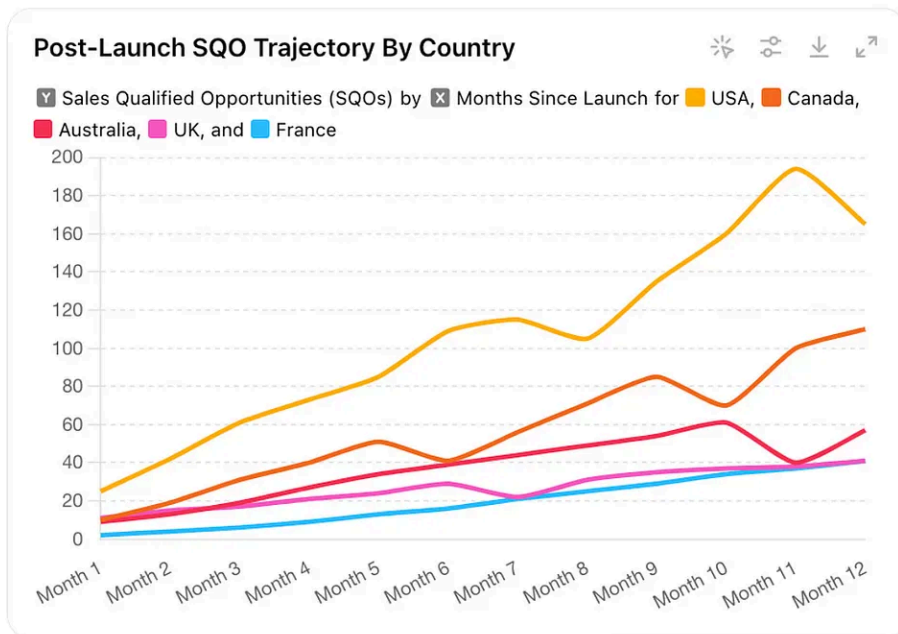
Sometimes it will also proactively visualize key trends; other times, you have to prompt it for that.

Here's how the conversation continued:



ChatGPT Output:

Can you plot the post-launch trajectory for the different markets? I find it difficult to parse the raw numbers



Here's the plot of the post-launch SQO trajectory for each country. This visual representation shows how the SQOs for France compare to the other markets over the first 12 months since their respective launches.

Screenshot from my third attempt; Image by author

It's nice that ChatGPT was able to pull the launch dates from the separate tab, and plot the performance for each country indexed by launch date.

We then got into seasonality; as you can see, I needed to provide the initial nudge and it took some back-and-forth, but ChatGPT did the work of identifying the correct patterns:



ChatGPT Output:

There seem to be some repeating seasonal pattern in markets like the US. What are the months where we see these month-over-month dips?



The month-over-month dips in the USA market occur in the following months:

- **Month 8:** A dip of -10 SQOs.
- **Month 12:** A dip of -29 SQOs.

These dips might indicate seasonal patterns or other external factors affecting the SQO numbers during these months. If you want to further analyze these patterns or adjust the forecasts for France to account for similar seasonal variations, please let me know! [?]

What actual calendar months are those? "Month 8" doesn't mean much to me since the data is indexed. Also, does this pattern match what we see in other markets?

[some detailed data cut for brevity; see transcript for full conversation]

Observations:

- **December:** Dips are observed in all markets during the month of December, indicating a potential seasonal impact.
- **August:** Both the USA and Canada experienced dips in August, suggesting another possible seasonal effect.

Screenshot from my [third attempt](#); Image by author

In several of my attempts, ChatGPT also gave a brief, but on-point summary of the Marketing mix, like this:



ChatGPT Output:

Marketing Channel	USA	Canada	Australia	UK	France
Organic	47.24%	19.90%	19.23%	17.50%	44.44%
Referrals	10.34%	2.49%	0.77%	3.75%	11.11%
Email	28.74%	49.75%	61.54%	62.50%	22.22%
LinkedIn	6.90%	19.90%	11.54%	10.00%	11.11%
SEM	6.78%	7.96%	6.92%	6.25%	11.11%

Observations

1. **Organic:** France has a high percentage of SQOs coming from organic channels (44.44%), which is more in line with the USA. This could indicate a strong content strategy or a high organic search volume.
2. **Referrals:** The referral rate for France is higher than that of Canada, Australia, and the UK but lower than the USA. This is within a reasonable range.
3. **Email:** The percentage of SQOs from email campaigns is significantly lower for France (22.22%) compared to other countries, which suggests either a lower focus on email marketing or different campaign effectiveness.
4. **LinkedIn and SEM:** The distribution for LinkedIn and SEM is relatively consistent with other countries.

Screenshot from my [fourth attempt](#); Image by author

I'd say this is overall okay.

It correctly highlights Organic and Email as key areas that deserve a closer look. Unfortunately, it also draws some odd conclusions; it doesn't make a lot of sense that Referrals in France would be as high as in the US given it's a new market without an established customer base.

Step 5: [Optional] Ask ChatGPT to re-do your work

If you used ChatGPT to sanity check a forecast, it sometimes offers to re-forecast based on your discussion.

In my case, I asked it to incorporate seasonality:



ChatGPT Output:

Conclusion:

There are consistent dips in SQOs around December across all markets, and another dip in August for North American markets. This suggests that seasonal factors such as holidays or vacation periods may impact SQO numbers during these months.

Would you like to adjust the forecast for France to account for these seasonal patterns, or do you need further analysis on these patterns? 🔄



Yeah, can you incorporate a modest seasonal dip for August and December in the France forecast as well?

The adjusted forecast for France now includes a modest seasonal dip of 10% in August (Month 3) and December (Month 7). The updated plot shows the adjusted trajectory for France alongside the original trajectories for other markets.

Here is a summary of the adjusted forecast for France:

Month	Original SQOs	Adjusted SQOs
1	2	2.0
2	4	4.0
3	6	5.4

Image by author

Final Thoughts

It sucks if you put a lot of effort into an analysis only for it to be torn apart by others. By sanity checking your work before you share it you can massively reduce the chance of that happening.

It's also a great way to build trust with more senior stakeholders and show that you are thinking like an executive.

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Right? Right??

okay I literally have this make a sound next to my ear



50

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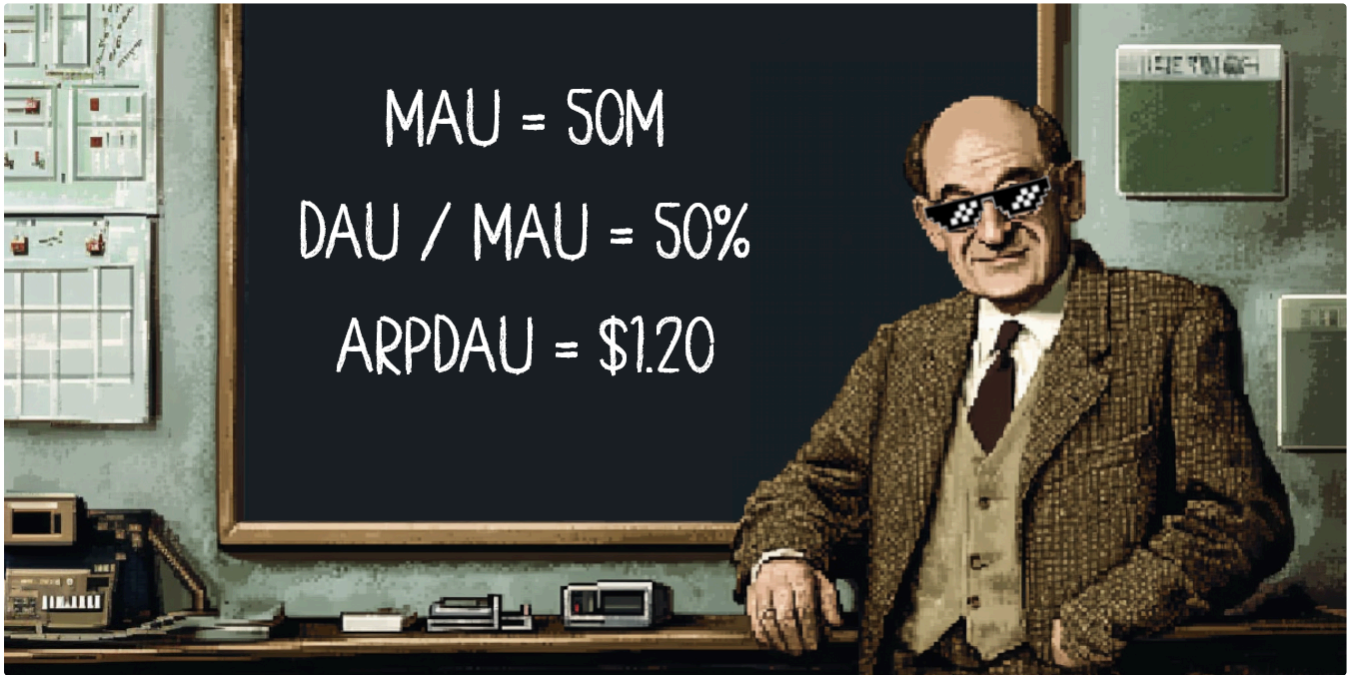
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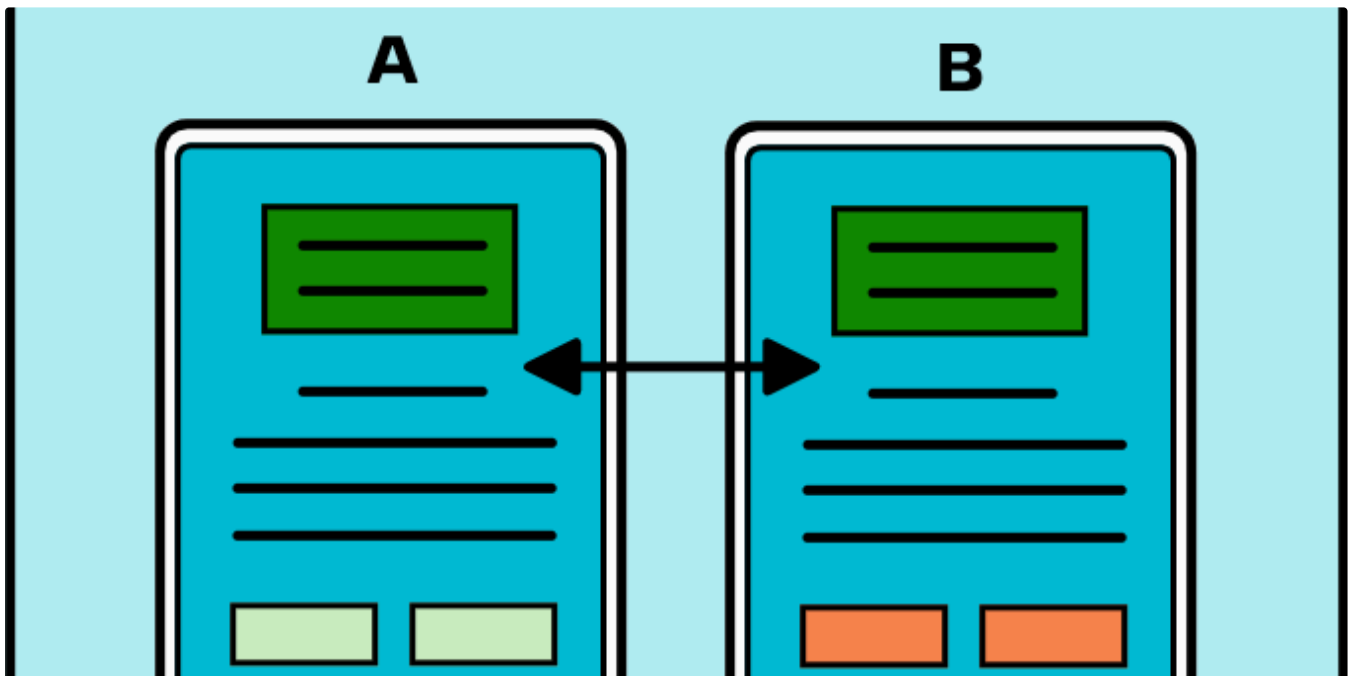
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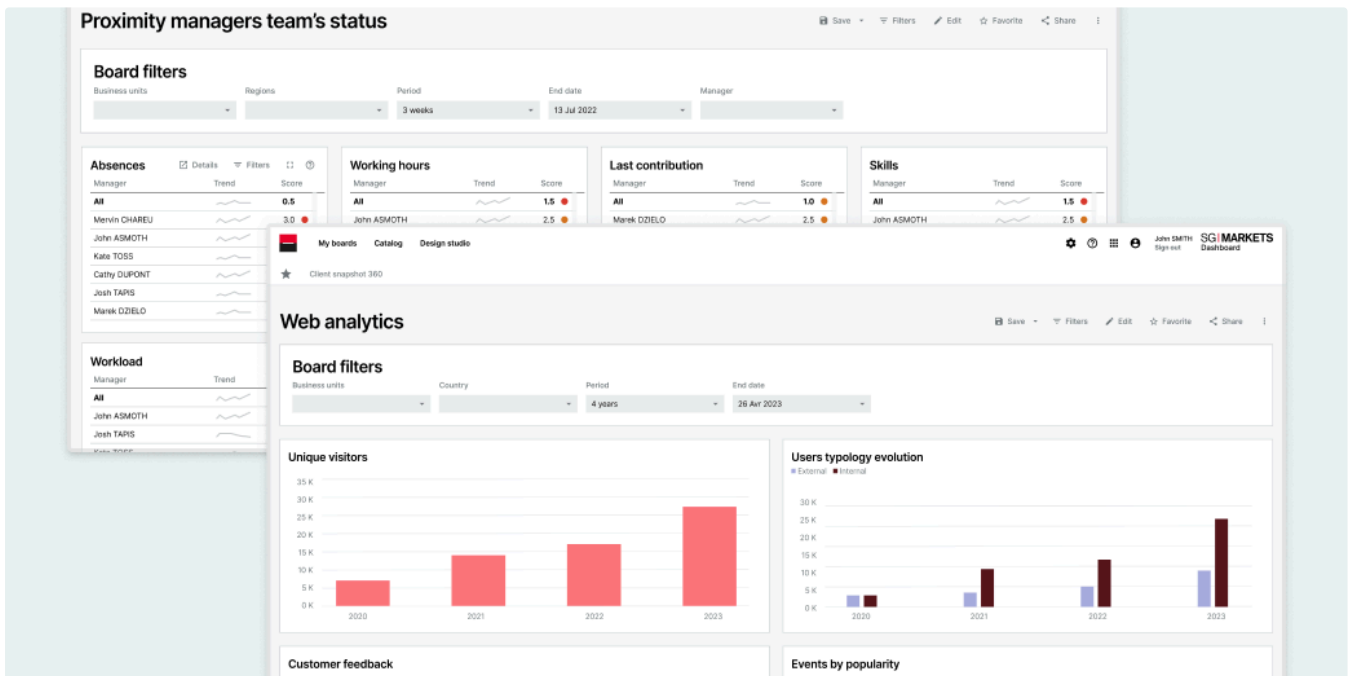
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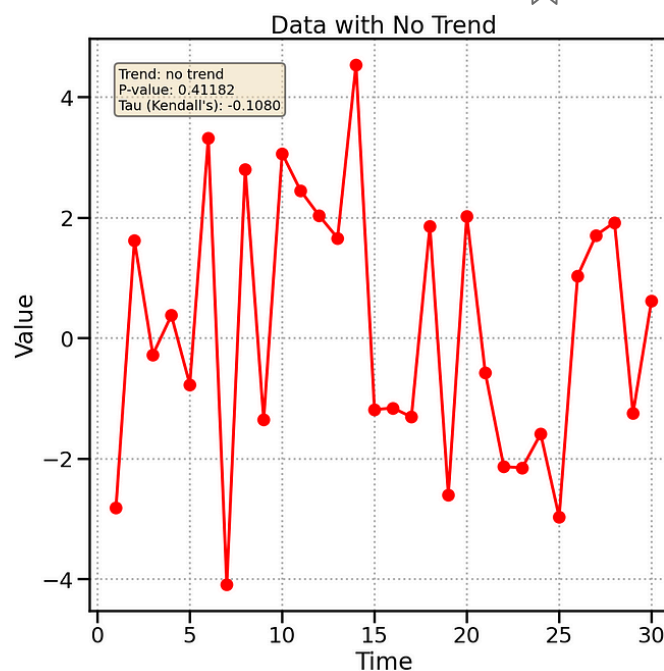
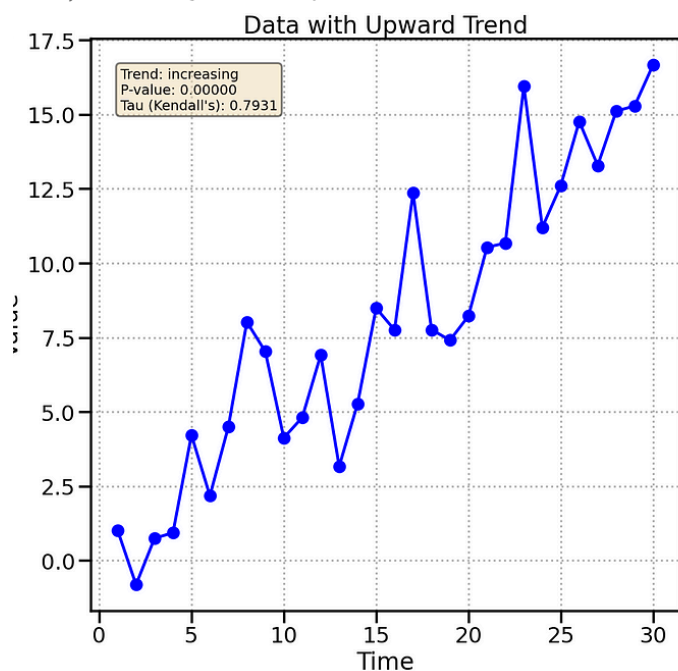


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